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enVisionMATH®

Common Core

Foundational Overview of the Common Core State Standards for Mathematics

Virtual Course

PARTICIPANT WORKBOOK

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Foundational Overview of the Common Core State Standards for Mathematics
Virtual Course
Participant Workbook

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Session 2: Teach Less, Learn More: The Standards for Mathematical Content	Self-Paced
Session 3: Developing Mathematical Habits of Mind: The Standards for Mathematical Practice	Self-Paced
Session 4: From Standards to Classroom Practice	Self-Paced
Session 5: Next-Generation Assessments	Live

Outcomes

At the conclusion of this Virtual Course, you will be able to

- identify the domains and concept categories included in the K–12 Standards for Mathematical Content;
- connect the K–12 Standards for Mathematical Practice to the National Council of Teachers of Mathematics' (NCTM) process standards and proficiencies as detailed in *Adding It Up: Helping Children Learn Mathematics*;
- identify ways to promote classroom discourse using the enVisionMATH Common Core curriculum that help students develop mathematical proficiency; and
- identify aspects of the Standards for Mathematical Practice that bring teaching with enVisionMATH Common Core closer to assessment.

Session 1: Getting to Know the Common Core State Standards for Mathematics

Big Questions

- What are the Common Core State Standards for Mathematics (CCSSM)?
- What are the goals of the CCSSM?

Overview of the CCSSM

List the two types of standards in the CCSSM:

1. Standards for Mathematical ___________________________
2. Standards for Mathematical ___________________________

Activity: Analyze Sample Assessment

Next-Generation Assessment Items

Assessment Item 1

Which number is both a factor of 100 and a multiple of 5?

A. 4

B. 40

C. 50

D. 500

(SBAC 2012b)

Assessment Item 2

Sort these four shapes. Use the characteristics labeled in the boxes below. Some shapes may belong in more than one box.

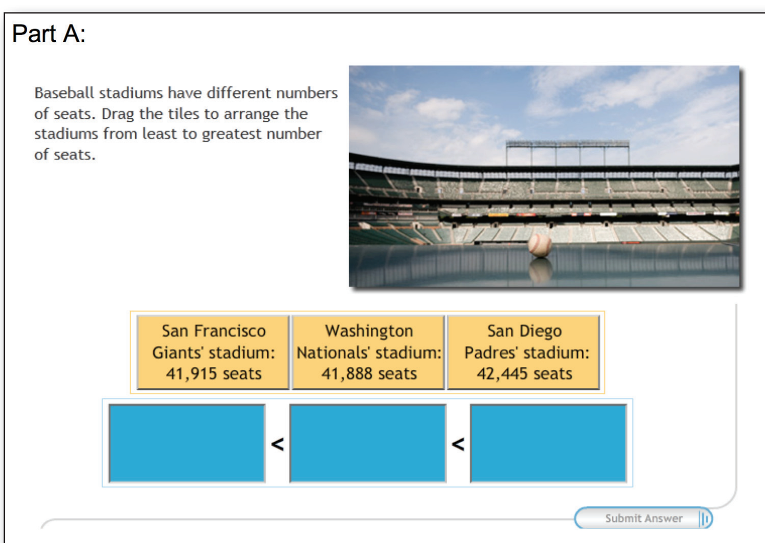
Rectangle | Rhombus | Right Triangle | Square

Click a shape and then click inside a box to place the shape in the box. Continue as many times as necessary.

Shapes with at Least One Right Angle	Shapes with at Least One Pair of Perpendicular Sides	Shapes with at Least One Pair of Parallel Sides

(SBAC 2012c)

Assessment Item 3



Part B:

Write your answer to the following problem in your answer booklet.

San Francisco Giants' stadium: 41,915 seats	Washington Nationals' stadium: 41,888 seats	San Diego Padres' stadium: 42,445 seats

Compare these statements from two students.

Jeff said, "I get the same number when I round all three numbers of seats in these stadiums."

Sara said, "When I round them, I get the same number for two of the stadiums but a *different* number for the other stadium."

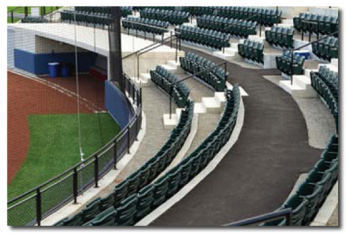
Can Jeff and Sara both be correct? Explain how you know.

Part C:

Write your answer to the following problem in your answer booklet.

When rounded to the nearest hundred, the number of seats in Aces Baseball Stadium is 9,100.

What is the greatest number of seats that could be in this stadium? Explain how you know.



(PARCC 2012)

Assessment Item 4 (Two of Four Total Tasks Provided)

Robot Maker—Task 1

You work for a company that makes robots. Your boss has asked you to design a new robot. The robot will contain a head, a body, 2 arms, and 2 legs. The first step is to draw what the front of your robot will look like. Use the practice grid paper provided to draw the front of your robot. Make sure to follow all of the guidelines below.

Guidelines:

1. The front of the body must be a rectangle with an area that is greater than 64 square centimeters but less than 140 square centimeters.
2. The front of the head must be a rectangle with a perimeter of 18 centimeters.
3. The front of each leg must be a quadrilateral that is not a rectangle.
4. The front of each arm must be a rectangle divided into equal parts with of the parts shaded.
34
5. Each eye must be shaped like a hexagon divided into equal parts with of the parts shaded.
13

The drawing must contain labels with any numbers and words that help your boss understand how you met each of the five guidelines. When you are sure your drawing is complete, copy your drawing to the answer sheet provided.

Robot Maker—Task 2

Your boss wants you to continue with your robot drawing. He has given you guidelines for creating a drawing of the back of the robot's head and body. Use the practice grid paper provided to draw the back of your robot's head and body. Make sure to follow **all** of the guidelines below.

Guidelines:

1. The back of the body must be the same size and shape as the front of the body.
2. The back of the body must contain a rectangular opening that is big enough to fit 2 AAA batteries placed side by side. The perimeter of this opening must be less than 16 centimeters. Use the batteries and a centimeter ruler to help you.
3. The back of the body must contain an on/off switch shaped like a rhombus. The rhombus must have a perimeter of 8 centimeters. Use a centimeter ruler to help you.
4. The back of the head must be the same size and shape as the front of the head.
5. A code is needed to lock and unlock the robot. Make this code by creating a skip-counting number pattern that starts with the number 7. The number pattern must have 5 terms. Write this code beneath the drawing of your robot.

The drawing must contain labels with any numbers and words that help your boss understand how you met the five guidelines. When you are sure your drawing is complete, copy your drawing to the answer sheet provided.

(SBAC 2012a)

Reflection

- What did you notice about the items?

____

____

____

- How are specific items similar to or different from current state assessment items?

____

____

____

- What are the implications for instruction?

____

____

____

Session 1 Assignment

Redefining Rigor for the 21st Century: Seven Survival Skills

Read the following descriptions from Tony Wagner's (2008) article, "Rigor Redefined," and identify the skills these employers value in prospective employees.

"Yesterday's answers won't solve today's problems."	
"Technology has allowed for virtual teams. We have teams working on major infrastructure projects that are all over the U.S. On other projects, you're working with people all around the world on solving a software problem. Every week they're on a variety of conference calls; they're doing Web casts; they're doing net meetings."	
"I can guarantee the job I hire someone to do will change or may not exist in the future, so this is why adaptability and learning skills are more important than technical skills."	
"I say to my employees, if you try five things and get all five of them right, you may be failing. If you try 10 things, and get eight of them right, you're a hero. You'll never be blamed for failing to reach a stretch goal, but you will be blamed for not trying."	
"[Young people] have difficulty being clear and concise; it's hard for them to create focus, energy, and passion around the points they want to make."	
"There is so much information available that it is almost too much, and if people aren't prepared to process the information effectively it almost freezes them in their steps."	
"For businesses it's no longer enough to create a product that's reasonably priced and adequately functional. It must also be beautiful, unique, and meaningful."	

Discussion Board

What are the characteristics of students who are college and career ready?

Session 2: Teach Less, Learn More: The Standards for Mathematical Content

Reflect on Session 1

Compare and contrast the list you posted on the Discussion Board that describes the characteristics of students who are college and career ready with the one provided in Session 2.

Are there any characteristics you would add to your own list?

Are there any characteristics you would add to the list provided? Why?

Big Questions

- How are the Standards for Mathematical Content organized?
- How do the Standards for Mathematical Content develop across the grade levels?
- How does enVisionMATH Common Core reflect the organization of the Standards for Mathematical Content?

Creating Better Standards

Watch and Respond

Phil Daro on the CCSSM
Chair, Mathematics College and Career Readiness Standards Work Group
Writing Team Mathematics K–12 Common Core Standards Committee
Senior Fellow, America's Choice

What is the significance of the connection between standards and performance?

Do you anticipate that the CCSSM will make your job easier or more difficult?

How would you summarize the concept of “teach less, learn more”?

Organization of the Standards for Mathematical Content

<table>
<tr><th>Grades K–2</th><th>Grades 3–5</th><th>Grades 6–8</th></tr>
<tr><td>Counting and Cardinality</td><td colspan="2"></td></tr>
<tr><td colspan="2">Operations and Algebraic Thinking</td><td>Expressions and Equations</td></tr>
<tr><td colspan="2">Number and Operations in Base Ten</td><td rowspan="2">The Number System</td></tr>
<tr><td></td><td>Number and Operations—Fraction</td></tr>
<tr><td colspan="2"></td><td>Ratios and Proportional Relationships</td></tr>
<tr><td colspan="2">Measurement and Data</td><td></td></tr>
<tr><td colspan="2"></td><td>Statistics and Probability</td></tr>
<tr><td colspan="3">Geometry</td></tr>
</table>

How to read the grade level standards

Standards define what students should understand and be able to do.

Clusters are groups of related standards. Note that standards from different clusters may sometimes be closely related, because mathematics is a connected subject.

Domains are larger groups of related standards. Standards from different domains may sometimes be closely related.

Domain

Number and Operations in Base Ten **3.NBT**

Use place value understanding and properties of operations to perform multi-digit arithmetic.

Standard

1. Use place value understanding to round whole numbers to the nearest 10 or 100.
2. Fluently add and subtract within 1000 using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction.
3. Multiply one-digit whole numbers by multiples of 10 in the range 10-90 (e.g., 9 × 80, 5 × 60) using strategies based on place value and properties of operations.

Cluster

(National Governors Association Center for Best Practices, Council of Chief State School Officers 2010, 5)

Pause and Reflect

Standards Sort Reflection

What did you notice?

____

____

____

What questions do you still have?

____

____

____

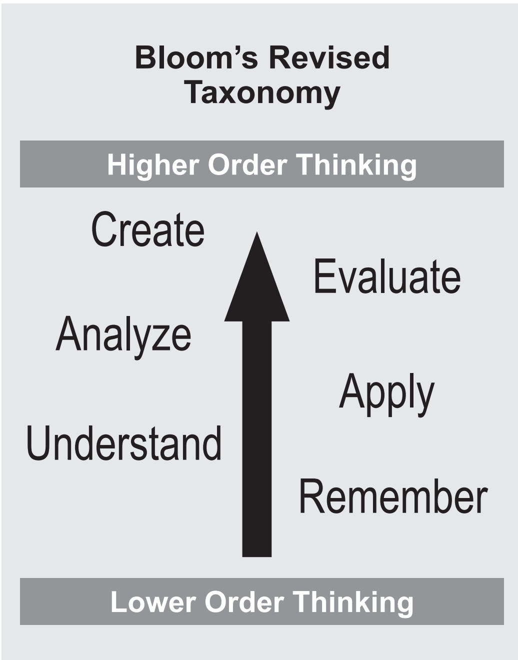
enVisionMATH Common Core and the Standards for Mathematical Content

Use the space below to record your notes on how enVisionMATH reflects the organization of the Standards for Mathematical Content.

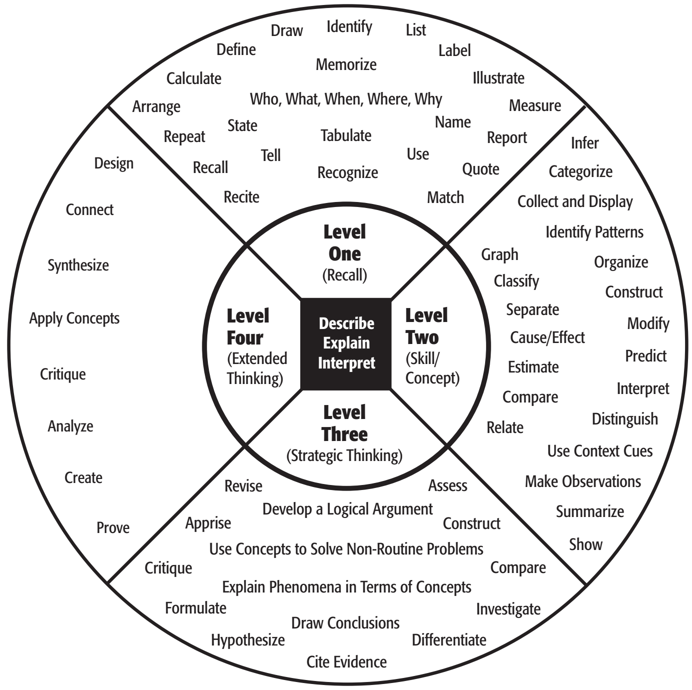
enVisionMATH Common Core and the Standards for Mathematical Content	Notes
Big Ideas	
Essential Understandings	
Content Organization	
Scope and Sequence	
Topic Teacher's Editions	
Each Lesson	

Cognitive Rigor

Bloom's Revised Taxonomy



Depth of Knowledge (DOK)



Level One

Recall: Activities require that students recall or recognize facts, information, concepts or procedures. Problems are one-step, well-defined, and require straight algorithmic procedures.

Level Two

Basic Application of Skill/Concept: Activities require that students use information or conceptual knowledge. They select appropriate procedures, follow two or more steps with decision points along the way, engage in routine problems, and organize or display data.

Level Three

Strategic Thinking: Activities require that students use reasoning and develop a plan or sequence of steps to approach the problem. They use decision making and provide justification, think abstractly and complexly, and understand that often more than one answer is possible.

Level Four

Extended Thinking: Activities require that students investigate or apply skills or concepts to real-world situations. They have time to research, think, process multiple conditions of the problem or task, and engage in non-routine manipulations that extend across disciplines or content areas and multiple sources.

(Hess 2006; Webb 2005; Webb 2006)

Cognitive Rigor Matrix

Revised Bloom's Taxonomy	Webb's DOK Level 1 Recall & Reproduction	Webb's DOK Level 2 Skills & Concepts	Webb's DOK Level 3 Strategic Thinking/ Reasoning	Webb's DOK Level 4 Extended Thinking
Remember Retrieve knowledge from long-term memory, recognize, recall, locate, identify	• Recall, observe, & recognize facts, principles, properties • Recall/ identify conversions among representations or numbers (e.g., customary and metric measures)			
Understand Construct meaning, clarify, paraphrase, represent, translate, illustrate, give examples, classify, categorize, summarize, generalize, infer a logical conclusion (such as from examples given), predict, compare/contrast, match like ideas, explain, construct models	• Evaluate an expression • Locate points on a grid or number on number line • Solve a one-step problem • Represent math relationships in words, pictures, or symbols • Read, write, compare decimals in scientific notation	• Specify and explain relationships (e.g., non-examples/examples; cause-effect) • Make and record observations • Explain steps followed • Summarize results or concepts • Make basic inferences or logical predictions from data/observations • Use models /diagrams to represent or explain mathematical concepts • Make and explain estimates	• Use concepts to solve non-routine problems • Explain, generalize, or connect ideas using supporting evidence • Make and justify conjectures • Explain thinking when more than one response is possible • Explain phenomena in terms of concepts	• Relate mathematical or scientific concepts to other content areas, other domains, or other concepts • Develop generalizations of the results obtained and the strategies used (from investigation or readings) and apply them to new problem situations
Apply Carry out or use a procedure in a given situation; carry out (apply to a familiar task), or use (apply) to an unfamiliar task	• Follow simple procedures (recipe-type directions) • Calculate, measure, apply a rule (e.g., rounding) • Apply algorithm or formula (e.g., area, perimeter) • Solve linear equations • Make conversions among representations or numbers, or within and between customary and metric measures	• Select a procedure according to criteria and perform it • Solve routine problem applying multiple concepts or decision points • Retrieve information from a table, graph, or figure and use it solve a problem requiring multiple steps • Translate between tables, graphs, words, and symbolic notations (e.g., graph data from a table) • Construct models given criteria	• Design investigation for a specific purpose or research question • Conduct a designed investigation • Use concepts to solve non-routine problems • Use & show reasoning, planning, and evidence • Translate between problem & symbolic notation when not a direct translation	• Select or devise approach among many alternatives to solve a problem • Conduct a project that specifies a problem, identifies solution paths, solves the problem, and reports results
Analyze Break into constituent parts, determine how parts relate, differentiate between relevant-irrelevant, distinguish, focus, select, organize, outline, find coherence, deconstruct	• Retrieve information from a table or graph to answer a question • Identify whether specific information is contained in graphic representations (e.g., table, graph, T-chart, diagram) • Identify a pattern/trend	• Categorize, classify materials, data, figures based on characteristics • Organize or order data • Compare/ contrast figures or data • Select appropriate graph and organize & display data • Interpret data from a simple graph • Extend a pattern	• Compare information within or across data sets or texts • Analyze and draw conclusions from data, citing evidence • Generalize a pattern • Interpret data from complex graph • Analyze similarities/differences between procedures or solutions	• Analyze multiple sources of evidence • Analyze complex/abstract themes • Gather, analyze, and evaluate information
Evaluate Make judgments based on criteria, check, detect inconsistencies or fallacies, judge, critique			• Cite evidence and develop a logical argument for concepts or solutions • Describe, compare, and contrast solution methods • Verify reasonableness of results	• Gather, analyze, & evaluate information to draw conclusions • Apply understanding in a novel way, provide argument or justification for the application
Create Reorganize elements into new patterns/ structures, generate, hypothesize, design, plan, construct, produce	• Brainstorm ideas, concepts, or perspectives related to a topic	• Generate conjectures or hypotheses based on observations or prior knowledge and experience	• Synthesize information within one data set, source, or text • Formulate an original problem given a situation • Develop a scientific/mathematical model for a complex situation	• Synthesize information across multiple sources or texts • Design a mathematical model to inform and solve a practical or abstract situation

(Hess 2009)

Pause and Reflect

Cognitive Rigor

How familiar are you with the DOK levels?

How could the Cognitive Rigor Matrix inform your instruction and assessment?

Learning Progressions

Select a standard at the grade level you teach from the Number and Operations in Base Ten (Grades 1–5) or The Number System (Grade 6) domain. Write it in the space provided below. Then, find the standard in the previous grade level that is related to this standard. Finally, determine the standard in the next grade level that is related.

Learning Progressions

Previous Grade Level Related Standard(s)	My Grade Level Standard	Next Grade Level Related Standard(s)

What do you notice?

Discussion Board

What resources are available to support your transition to the CCSSM on your state's Department of Education Web site?

List at least two content changes that have occurred at your grade level when you compare the CCSSM to your previous state standards.

Session 3: Developing Mathematical Habits of Mind: The Standards for Mathematical Practice

Big Questions

- How will the Standards for Mathematical Practice inform instruction?
- How does enVisionMATH Common Core support the Standards for Mathematical Practice?

Eight Standards for Mathematical Practice

1. Make sense of problems and persevere in solving them.
2. Reason abstractly and quantitatively.
3. Construct viable arguments and critique the reasoning of others.
4. Model with mathematics.
5. Use appropriate tools strategically.
6. Attend to precision.
7. Look for and make use of structure.
8. Look for and express regularity in repeated reasoning.

(National Governors Association Center for Best Practices, Council of Chief State School Officers 2010, 6–8)

Overarching Habits of Mind of a Productive Mathematical Thinker 1. Make sense of problems and persevere in solving them. 6. Attend to precision.		
Reasoning and Explaining	**Modeling and Using Tools**	**Seeing Structure and Generalizing**
2. Reason abstractly an quantitatively. 3. Construct viable arguments and critique the reasoning of others.	4. Model with mathematics. 5. Use appropriate tools strategically.	7. Look for and make use of struture. 8. Look for and express regularity in repeated reasoning.

(McCallum 2011)

Long-Standing Importance

Watch and Respond

Juanita Copley on Adding It Up
enVisionMATH Author
Chair, Curriculum and Instruction Department
College of Education University of Houston
Houston, Texas

What are the five areas that make up mathematical proficiency as described in the *Adding It Up: Helping Children Learn Mathematics* publication?

Why is it important for students to have procedural fluency *and* conceptual development?

Model with Mathematics: A Closer Look

What do you think of when you hear the word *modeling*?

Pause and Reflect

What does it mean to model with mathematics?

Does the description found on page 7 of the CCSSM document sound similar to or different from your initial thinking? How so?

What are some of the key phrases that illustrate students' habits of mind? What would you expect to see if students demonstrate modeling with mathematics?

enVisionMATH Common Core and the Standards for Mathematical Practice

enVisionMATH Common Core and the Standards for Mathematical Practice	Notes
Problem-Based Interactive Learning	
Visual Learning Bridge	
Practice Problems	

Pause and Reflect

A Closer Look at a Mathematical Practice in enVisionMATH Common Core

Part 1

You took a closer look at the Standard for Mathematical Practice, “Model with mathematics” (National Governors Association Center for Best Practices, Council of Chief State School Officers 2010, 7). Select another Standard for Mathematical Practice that you would like to better understand. Use the CCSSM document or pages 24–27 in your *Teacher’s Program Overview* to read the description of the Standard for Mathematical Practice, and then paraphrase it below.

Part 2

Find two places in enVisionMATH Common Core where this particular Standard for Mathematical Practice is highlighted. What other Standards for Mathematical Practice are highlighted?

Understanding Children's Thinking

Grade	CCSSM
5	**Use equivalent fractions as a strategy to add and subtract fractions.** Solve word problems involving addition and subtraction of fractions referring to the same whole, including cases of unlike denominators, e.g., by using visual fraction models or equations to represent the problem. Use benchmark fractions and number sense of fractions to estimate mentally and assess the reasonableness of answers.

(National Governors Association Center for Best Practices, Council of Chief State School Officers 2010, 36)

Example: Fair Share

There is a delicious cookie that four children at a party want to share.

A. Show how the four children might share the cookie. How much does each person get?

B. Just before the cookie is handed out, one child is picked up from the party without eating or taking her piece. The three people remaining want to share the cookie. How could they do that fairly? How much does each person get?

Watch and Respond

As you watch the video, examine Felisha's work and explain her mathematical thinking during each step of the process.

A. Sharing cookie with 4 children

B. Sharing cut cookie with 3 children

C. Showing how much each child gets

D. Recording the amount each child gets

Without the context of the children sharing a cookie, how do you think Felisha would do when asked to find the sum of 1/4 and 1/12?

Place a check mark by the Standards for Mathematical Practice that you observed as Felisha worked on the problem.

___ Make sense of problems and persevere in solving them.
___ Reason abstractly and quantitatively.
___ Construct viable arguments and critique the reasoning of others.
___ Model with mathematics.
___ Use appropriate tools strategically.
___ Attend to precision.
___ Look for and make use of structure.
___ Look for and express regularity in repeated reasoning.

(National Governors Association Center for Best Practices, Council of Chief State School Officers 2010, 6–8)

Podcast

Classrooms that Promote Understanding

Nature of Classroom Tasks

Role of the Teacher

Social Culture of the Classroom

Mathematical Tools

Equitable Opportunities for All Students

Discussion Board

Review the eight Standards for Mathematical Practice, and describe specific ways that you can develop a classroom culture that provides opportunities for students to develop these habits of mind.

Session 4: From Standards to Classroom Practice

Reflect on Session 3

Discussion Board: A Culture of Thinking and Learning

Read through the Discussion Board posts from Session 3. What are some ideas from other Virtual Course participants that you can add to your own list of specific ways that you can develop a classroom culture that provides opportunities for students to develop these habits of mind?

Big Questions

- What kinds of learning tasks allow students the opportunity to demonstrate their mathematical proficiency?
- How can you establish a classroom culture of thinking and learning that supports the vision of the CCSSM?
- How can you utilize enVisionMATH Common Core to implement the CCSSM?

A Problem-Based Approach to Mathematics Instruction

Three-Part Lesson Format

BEFORE	
What I do:	What students do:
DURING	
What I do:	What students do:
AFTER	
What I do:	What students do:

Problem-Based Interactive Learning in enVisionMATH Common Core

Engage Students	Students Solve and Discuss a Problem	Make the Important Mathematics Explicit	Deepen Understanding
Begin with Engage, which is done with the entire class to make connections. • Set the Purpose connects the new concept that students will learn to previously learned concepts. • Connect relates the new concept that students will learn to something in the real world.	Pose the problem. • The problem will require some thinking. • Students will be able to solve many problems in more than one way. Allow time for students to solve the problem. • Students usually work in pairs or groups during lessons. • The teacher observes and facilitates as students work. Have students share their thinking and work. • Students explain their thinking and work. • Students share alternative solutions or ways of thinking.	End by making the important mathematics explicit by connecting to the students' thinking and work.	As time permits, provide the Extend feature to increase knowledge and depth of understanding.

Pause and Reflect

Problem-Based Interactive Learning

How does Problem-Based Interactive Learning reflect a problem-based approach to instruction?

A Problem or Learning Task Defined

A *problem* can be defined as an activity or task that encourages the learning of mathematics by

- beginning where the students are;
- highlighting the mathematics that students are to learn in a problematic or engaging way; and
- requiring justifications and explanations for answers and methods.

(Van de Walle and Lovin 2006, 11–12)

Task Analysis

Analyze three different problems or tasks students might encounter in Problem-Based Interactive Learning, formative assessment, or on a future assessment aligned to the CCSSM. Use the following questions to guide your analysis.

	What important mathematics does this problem or task address?	How does each task or problem provide opportunities for students to justify or explain their thinking?
Problem-Based Interactive Learning		
Formative Assessment		
Future Assessment Aligned to the CCSSM		

Pause and Reflect

What did you notice about the three different problems or tasks?

How are they similar? How are they different?

Classroom Discourse

Classroom discourse is about classroom conversations that are grounded on mathematical understandings.

Teachers that encourage rich mathematical conversations

- choose high-level learning tasks;
- encourage students to work independently of the teacher, either individually or cooperatively in groups;
- encourage students to analyze situations and pose higher-order questions;
- listen carefully to students' approaches;
- use questioning to clarify misconceptions, extend thinking, and encourage alternative methods; and
- facilitate whole-group conversations focused on the important mathematics or essential understandings.

Watch and Respond

John Van de Walle on Approach to Teaching
Professor Emeritus, Virginia Commonwealth University
Author, *Elementary and Middle School Mathematics: Teaching Developmentally*

What are the seven "tough ideas and difficult challenges" faced by teachers when they teach with problems in a way that promotes understanding?

1.

2.

3.

4. ______________________________

____

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5. ______________________________

____

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6. ______________________________

____

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7. ______________________________

____

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Problem-Based Interactive Learning and the Mathematical Practices

Identify two Problem-Based Interactive Learning activities in your Teacher's Edition (Part 2 of each lesson), and discuss how each one facilitates the development of the highlighted Standard for Mathematical Practice. Additionally, identify other Standards for Mathematical Practice that you can incorporate in each Problem-Based Interactive Learning activity.

Lesson and Page	Overview of the Problem	Highlighted Standard for Mathematical Practice	Other Standards for Mathematical Practice

Podcast

Formative Assessment

1. How would you describe formative assessment?

2. What are the four essential elements of the formative assessment process?

Session 5: Next-Generation Assessments

Big Questions

- How will the CCSSM change assessment?
- What assessment opportunities does enVisionMATH Common Core provide that support new changes to assessment?

Performance Tasks Defined

An effective performance task

- represents content that is meaningful to students and involves authentic, relevant tasks that simulate college, career, or life applications;
- requires students to engage in critical thinking, analytic reasoning, and problem solving;
- addresses multiple standards that require an integration of mathematical concepts and skills;
- may range in difficulty, but typically includes multiple embedded steps that lead to a final activity;
- elicits direct observable evidence of performance; and
- supports students in developing the ability to transfer these skills to new situations.

enVisionMATH Common Core Assessment Resources

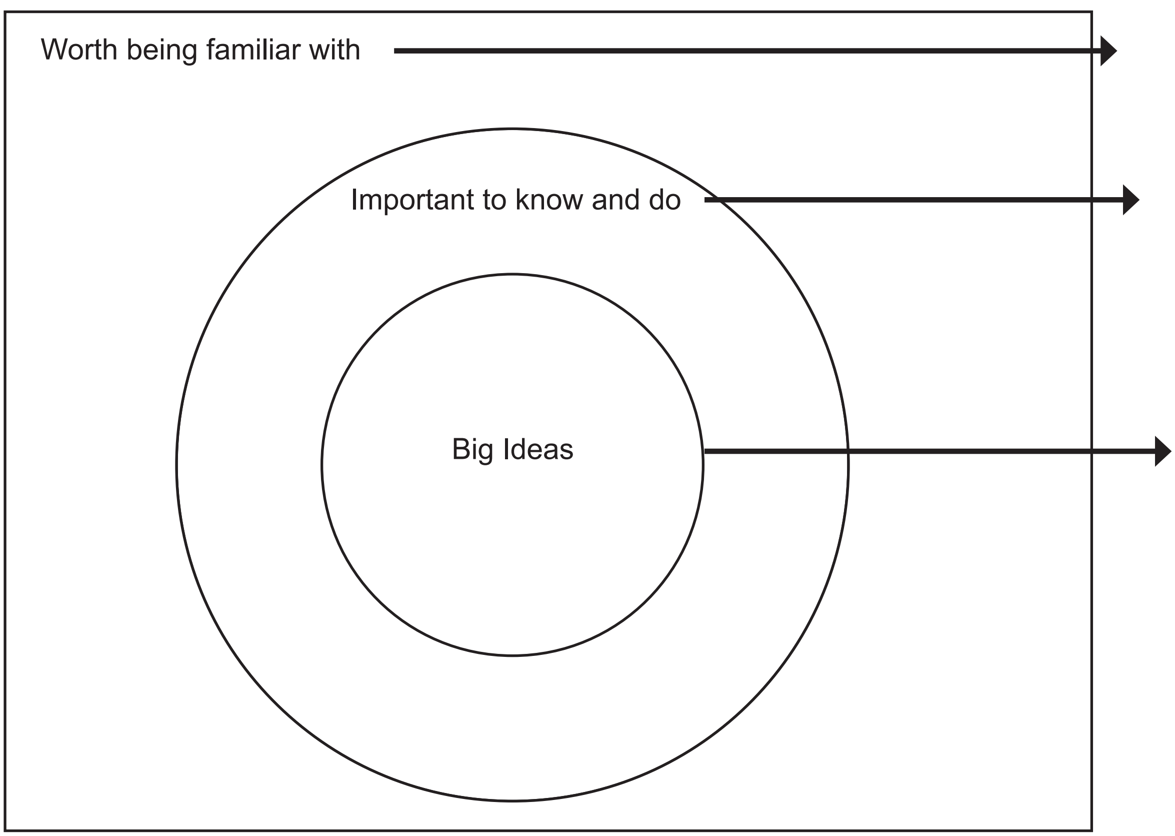
Reflecting on Implications for Implementation

Reflect on what you have learned in this Virtual Course:

1. What are some things worth being familiar with?

2. At this point, what is most important for you to know and do?

3. What are the big ideas for change in your own classroom?



(Wiggins and McTighe 2005, 71)

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Appendix

Watch and Respond

Juanita Copley on Adding It Up

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1. The five areas that make up mathematical proficiency as described in the *Adding It Up: Helping Children Learn Mathematics* publication are procedural fluency, conceptual development, adaptive reasoning, strategic competence, and productive disposition.
2. It is not enough for students just to know how to complete a procedure; students must understand the concept behind procedures. Juanita Copley illustrates this in the example she provides of students being able to count by fives, but not understanding the concept of what counting by fives actually means.

Watch and Respond

John Van de Walle on Approach to Teaching

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1. Accept that true growth comes from confronting difficulty rather than being given solutions.
2. Require student to make sense of mathematics.
3. Focus on big ideas.
4. Require students to give more than answers.
5. Allow students to struggle, and provide support rather than answers.
6. Do not “show and tell.”
7. Support student communication, but do not speak for students.